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Working memory

Strategies for classroom support

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WHAT IS WORKING MEMORY?

Working memory is the ability to hold and manipulate information in the mind for brief periods of time in the course of ongoing mental activities. It consists of three sub-components:

- Verbal short-term memory ⟶ stores verbal information
- Visual-spatial short-term memory ⟶ stores information relating to vision, space or movement
- The central executive ⟶ controls attention and coordinates activity both within the working memory system and between working memory and other cognitive systems such as long-term memory (Alloway & Gathercole, 2008, pp11-12).

Working memory is limited in capacity. The capacity varies between individuals and is affected by characteristics of the material that is being stored. Information is lost from working memory when we are distracted or its limited capacity is overloaded. Short-term memory involves storage, whereas working memory is involved more generally when a task involves storing and mentally manipulating information.

Working memory capacity increases from childhood through to adolescence, when adult levels are reached. Within a particular age group, there is a wide variation in working memory capacity between individuals. Children typically start to use rehearsal in verbal short-term memory at about seven years of age, and at this age also shift towards preferring to remember information in terms of verbal characteristics if possible. Increases in working memory capacity with age relate to improvements in the efficiency of processing and attention.

Children's progress in reading, mathematics and science is closely related to their working memory capacities, across the full range of school years. Children with learning difficulties in reading and mathematics typically have very poor working memory capacities. Poor working memory performance does not appear to be due to more general factors such as low IQ or language difficulties. The poor rates of learning in children with low working memory capacities are due in large part to memory overload in structured learning activities, which causes them to forget crucial information and so to fail in these tasks.

WHY ASSESS WORKING MEMORY?

Working memory is a critical factor in predicting the ability to learn. We need to consider working memory within the learner's profile because:

- the learner may have limited working memory capacity (below 10th percentile) which is unlikely to improve, therefore adjustments will be needed in the classroom; or
- the learner may not be utilizing all of their working memory capacity, therefore understanding the learner's particular profile will enable effective strategies to be adopted.

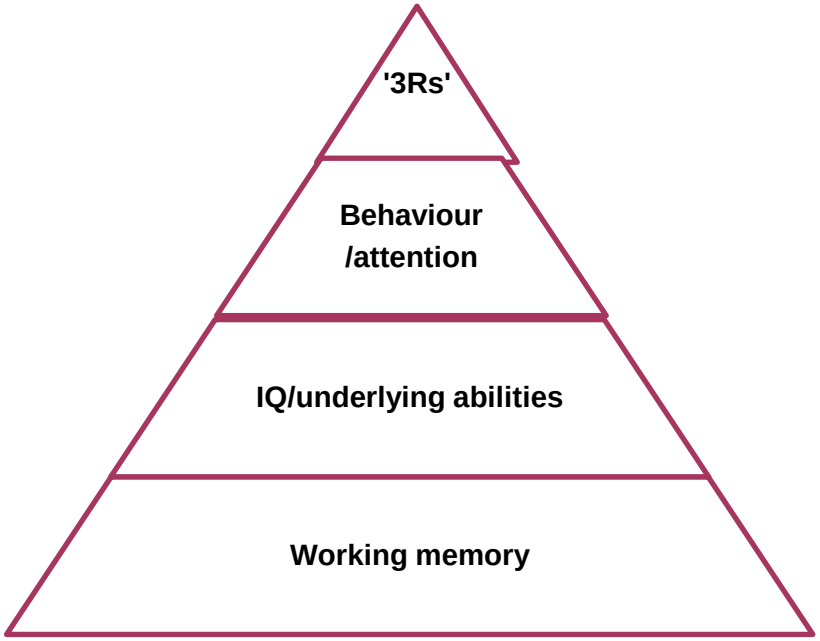
Any degree of difficulty with memory, attention and/or processing will prevent full access to the curriculum. A working memory friendly classroom will:

- reduce memory overload, irrespective of cause;
- train in compensatory techniques;
- monitor for transference of strategies learnt; and
- foster independent skills.

THE LEARNING PYRAMID

Tracy Packiam Alloway argues that working memory is the foundation to all learning – interventions based on the '3Rs' will have no discernable impact on learning outcomes if there are weaknesses in working memory. Many with a high IQ do not do as well in academic terms as expected because of weaknesses in their working memory and learners with working memory problems can be misdiagnosed as lazy or unmotivated (Packiam, 2011, pp10-11).

Unless compromised working memory is addressed, Wave 3 interventions are unlikely to succeed.



HOW IS WORKING MEMORY ASSESSED?

Working memory will be only one facet of an assessment – and as with any assessment, collation of background information will be invaluable in building up a complete profile of a learner. This may include the following considerations:

- Relevant pre-school history
  - was the learner born prematurely?
  - were developmental milestones reached within the usual time scales?
  - any input from speech and language therapy?
  - is there any history of glue ear, tonsil infection, adenoids etc?
- Does there appear to be visual discomfort or tracking problems?
- Has eyesight been tested in the last 18 months?
- Does the learner need a hearing test?
- If letter and number formation remain large and poorly formed/orientated
  - what are other fine motor skills like?
  - is the learner not yet sure which hand to write/draw with?
  - is there any difficulty with gross motor skills?
- Is there any family history of specific learning difficulties that might be significant?
- Any other disabilities or health issues to be aware of (eg epilepsy)?

Should an assessment of working memory be needed, a staged approach might be:

1. Information gained through a consultation
2. Observation (by the school) based on the Working Memory Rating Scale (see below) or similar
3. Screening for working memory
4. In depth assessment

OVERVIEW OF ASSESSMENT TOOLS

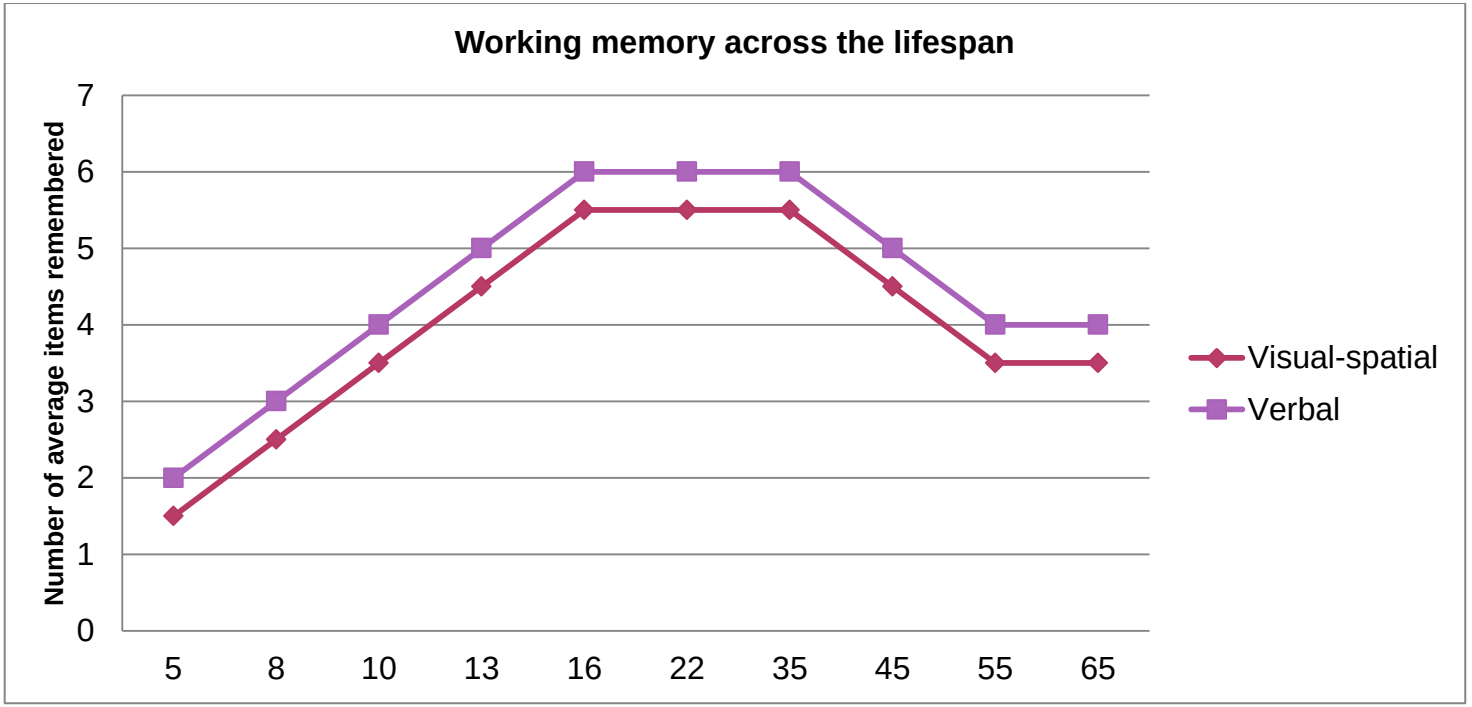
Name of test	Publisher/Cost [correct as of Feb 2011 excluding VAT unless stated]	Description/Possible uses
Digit span testing	Turner & Risdale's Digit Memory Test (revised 2004) is available to download online: www.dyslexiaaction.org.uk/copyfreetests	Digit span testing is a measure of memory span which is the number of items that a person can retain and recall. It can be seen as a measure of working memory (or short-term memory, depending on the psychological framework used), although other factors such as attention and comprehension also contribute to the performance on this test. In a typical test of memory span, a list of random numbers is read out at about the rate of one per second. The test begins with two to three numbers, increasing until the person commits errors. Recognisable patterns (eg 2, 4, 6, 8) should be avoided. At the end of a sequence, the person being tested is asked to recall the items in order. Reverse repetition is more difficult and requires more processes besides immediate recall.
'Working Memory Index' in *Wechsler Intelligence Scale for Children 4th UK Ed* (WISC-IV UK) – David Wechsler	Pearson Education (2004) Complete kit: £9.48.50 ISBN: 9770749123659 www.psychcorp.co.uk	The *WISC* is a standardised assessment of cognitive abilities in learners aged six to 16 years 11 months. The 'Working Memory Index' comprises digit span, letter-number sequencing and arithmetic. **NB:** Test requires professional post graduate qualification and training in clinical or educational psychology.
Working Memory Rating Scale (WMRS) – Susan	Pearson Education (2008) Complete kit: £40.00	The *WMRS* is a behavioural rating scale developed to facilitate easy identification of working memory deficits in learners aged five to 11 years by classroom teachers. The WMRS consists of 20

Gathercole, Tracy Packiam Alloway & Hannah Kirkwood	ISBN: 9780749151003 www.psychcorp.co.uk	short descriptions of problem behaviours that differentiate learners with low and average working memory abilities. The teacher rates how typical each behaviour is of the learner on a scale ranging from *not typical at all* to *very typical*. It provides a systematic evaluation of the potential working memory problems that can be provided by observation alone. Identification can then be followed up by an examination of the individual's detailed profile of working memory strengths and weaknesses using standardised measures such as the Automated Working Memory Assessment (AWMA).
Automated Working Memory Assessment (AWMA) – Tracy Packiam Alloway	Pearson Education (2007) Complete kit: £191.50 (provides two installations) ISBN:9780749141950 www.psychcorp.co.uk	The *AWMA* is a PC-based assessment of working memory skills for learners aged four to 22 years. The testing sequence is pre-set, test scores are calculated by the computer program and an interpretation of how working memory scores will affect learning is provided. There are three levels of assessment within AWMA: *AWMA Screener* This is made up of two tests and is suitable for screening learners with suspected working memory difficulties. *AWMA: Short Form (AWMA-S)* This consists of four tests and is recommended to screen learners who are suspected to have memory difficulties, but the specific area of their difficulties is not known. *AWMA: Long Form (AWMA-L)* This consists of all 12 tests and is recommended for confirmation of significant working memory problems for learners identified as having working memory problems in the classroom. **NB:** Not compatible with Windows 7
Working Memory Test Battery for	Pearson Education (2001)	The *WMTB-C* provides an assessment of working memory in five to 15 year olds. The test is designed to reflect the three component structure of the Working Memory Model proposed by Baddeley and

Children (WMTB-C) – Sue Pickering and Sue Gathercole	Complete kit: £312.00 ISBN:9780749120207 www.psychcorp.co.uk	Hitch. The specialised components each deal with different aspects of working memory. They include the Central Executive (CE), which is involved in the control and regulation of the Working Memory System and two 'slave systems.' One slave system is known as the Phonological Loop (PL) and is responsible for holding verbal information for short periods; the other is the Visuo-spatial Sketchpad (VSSP) which holds information in visual and spatial form. Research indicates that the *WMTB-C* is useful in identifying learner who perform poorly at school, including children with specific learning difficulties such as dyslexia. Data is also available about the extent to which the *WMTB-C* predicts achievement at school more directly, by exploring the profiles of learners at Key Stage 1, Key Stage 2 and Key Stage 3 using school-based assessments of progress on the National Curriculum. The *WMTB-C* has been validated against existing well established tests of achievement, including the British Picture Vocabulary Scale, subtests of the British Ability Scales, Neale Analysis of Reading Ability, Group Arithmetic test and subtests of the Differential Abilities Scales. **NB:** This product will be going out of print when current stock is sold but record forms will continue to be provided for two years from January 2011.

WORKING MEMORY AND LEARNING

The most dramatic growth in terms of how much the average individual can remember is during childhood. These years are crucial as working memory increases more in the first ten years than it does over the lifespan. Working memory reaches a peak and plateaus during the twenties before it begins to decline. Working memory is a relatively stable construct and all working memory components are in place by four years of age. While working memory does increase with age, its relative capacity remains constant. In the classroom, verbal working memory is used to remember instructions, learn language and perform comprehension tasks. Visual-spatial working memory is used to remember sequences of events, patterns, images and mathematical skills (Packiam, 2011, pp2-4).



LIMITATIONS TO WORKING MEMORY

There are three main limitations to working memory (Packiam, 2011, pp4-7).

Space

Up until age 15, working memory is constantly increasing in size. But some people's working memory grows faster than others, while others' working memory grows more slowly.

Time

When a teacher speaks too quickly, a learner with poor working memory is unable to process all the information fast enough and will invariably give up on the activity. Activities that take a long time to process can overburden a learner with poor working memory. Time pressure can also make a manageable task too demanding.

Effort

Difficulty keeping focus is a key feature of working memory overload. Learners with poor working memory have to use a lot of effort when they have to juggle two or more things.

RECOGNISING POOR WORKING MEMORY CAPACITY AND MEMORY OVERLOAD IN THE CLASSROOM

Common signs of poor working memory capacity and memory overload include some or all of the following:

- poor attention and listening skills
- easily distracted
- restlessness
- short concentration span
- day dreaming or 'zoning out'.

PRACTICAL APPLICATIONS

Below are some of the characteristic behaviours listed in the *Working Memory Rating Scale* (WMRS) which may be seen in the classroom of learners with poor working memory capacity alongside suggested strategies. Page references are to *Working Memory & Learning: A Practical Guide for Teachers* by Susan E Gathercole and Tracy Packiam Alloway (2008).

COMPLETING TASKS INDEPENDENTLY		
Classroom behaviour	**Links to working memory**	**Strategies**
To move on to the next step in an activity, needs frequent prompts by teaching staff	• The learner may struggle to remember several steps of a task if instructions for a task are only given verbally • Lengthy tasks with new or unfamiliar content may overload the working memory • Lack of prior knowledge places additional demands on working memory	• Simplify tasks – break down into separate independent steps (p78) • Use visual prompts • Use simple checklists or a flow chart to identify each step in a sequence • Use diagrams with key elements of task depicted by colour or picture cue • Review each step as completed, adjusting demands of task if necessary • Prompt with questions to keep focus on task, eg 'What sentence are you going to write?' 'Which word is missing to complete your sentence?' • Link steps with familiar or meaningful experiences – rather than introducing topic to whole class and then taking out learners who need additional explanation, sometimes it would be beneficial to 'pre-teach' a topic before it is introduced to the whole class • Parents/carers can also be cued in to class topics

		to help prepare learners in advance
Frequently asks for help	• Sign of poor working memory capacity (p71) • Few strategies for independent working • Working memory overload in the task	• More support in planning stage – check understanding and break down into smaller steps • Use spidergram to gather ideas and number points • Correct topic word spellings before drafting • Develop learner's range of spelling strategies, eg highlighting letter patterns, use personal dictionary, target high frequency words on a treasury tag, mnemonics, rhymes, rules • Place with a 'memory buddy'
Abandons activities before completion – the learner gives up, forgetting what to do, sometimes switching attention to another activity or distracting others	• Common consequence of working memory failure	• Monitor and pre-empt by asking the learner what s/he is doing and what s/he intends to do next • Task may need to be repeated and possibly modified (p75) • Encourage learner to request repetition of information/instruction • 'Say it back' – the learner repeats the instructions or talks through steps of activity. This helps the learner to recall and strengthen her/his grasp of a task • Partner learner with a classmate who has good working memory skills (p80)
Frequently stops during lengthy activities or those involving multiple steps - loses place in complicated	• Forgets some or all of information required to complete task or activity • Cannot remember and so does not	• Break down task into small steps, each achieved before moving on to next step • Intersperse a demanding task with an easier,

activities	follow multi-step instructions (p70) • Cannot remember the words in a sentence s/he is attempting to write (p70) • Low working memory affects ability to carry out multi-step instructions (p70)	procedural one • Allow more time • Reduce the amount of writing required by offering alternative methods of recording, eg filming, audio recording, working with a partner who scribes • New information should correspond to previously acquired knowledge • Review topic before activity (p76)
Forgets how to continue an activity that was previously started, despite teacher explanation	• The learner needs to be 'cued in' and may struggle to take in an explanation that relies on easy retrieval of prior knowledge	• Work on previous example together or link with a 'buddy' until the learner gradually becomes more independent • Be prepared to repeat the task, to familiarize with the aims, modifying and simplifying the language of instructions
Benefits from continued teacher support during lengthy activities, especially written tasks	• Too many steps to the task to retain in memory • Learner may not know how to use writing aids to support memory • Difficulties in the past may have led to 'learned helplessness'	• Encourage learner's independence through awareness of preferred learning styles and strategies to support memory • Try Clicker 5 program for writing support to reduce working memory load - pre-enter topic words - pre-enter high frequency words • Use word prediction, eg Co-writer • Use writing frames
Not able to focus during	• Concentration may be affected by	• Create a supportive physical environment – consider seating in relation to teacher

activities	demands of task causing working memory overload	• Reduce distraction with a work station for complex tasks • Use picture prompts or sequence of photographs to match activity plan • Use clear visual images/uncluttered layout on task sheet • Draw on learner's long-term memory of personal experiences to make the activity meaningful
Depends on neighbour to remind them of the current task	• Not actively listening when the instructions are given • Learner is masking problems with understanding or memory overload • Difficulties in the past may have led to 'learned helplessness'	• Ensure that the learner is ready and listening to instructions – minimize distraction by alternative seating • Encourage learner to use simple step guide to build independence and confidence • Jot down important notes on a desk top whiteboard • Note down a number, spelling or use a picture cue as a reminder

DIFFICULTIES REMEMBERING OR PROCESSING SPOKEN INFORMATION		
Classroom behaviour	**Links to working memory**	**Strategies**
Puts hand up to answer a question but forgets what s/he intended to say when asked	• Difficulty retrieving and producing spoken information • Difficulty holding on to thoughts under pressure • Easily distracted by what others say while waiting for her/his turn to speak • Losing track of an argument or stages in an experiment	• Ask the learner straight away before s/he forgets • Repeat the question or previous point made in class discussion to give learner time to answer • Forewarn when the learner will be asked to answer so that s/he has time to prepare • Help by jotting key ideas down on whiteboard – flow chart with key ideas, pictures cues, mind mapping, timelines etc • Ask a teaching assistant to draw the picture cues while class teacher addresses the class
Does not respond or is reluctant to answer when asked direct questions	• Questions may be too lengthy, grammatically complex or unfamiliar • Difficulty processing verbal information – weak memory for information just heard • Cannot recall and order ideas under pressure • Expectation of producing clear and concise answer becomes a barrier	• Use name first to gain attention • Ask question and allow time for response, eg count the seconds it takes the learner to process the instruction/question and adjust the way you ask the question, slowing speech and reducing words • Simplify structure of sentences (p77) • Tailor specific questions to learner • Ask learner to repeat instructions to check understanding
Does not follow classroom instructions accurately, eg carries out some but not all	• Learner may not be ready to listen to instructions	• Try two step instructions, using demonstration and non-verbal cues for extra emphasis

steps in an instruction	• Easily distracted • Too many instructions given at once	• Give instructions with a simple grammatical structure and meaningful or familiar content
Raises hand but gives inappropriate or incorrect answers	• Weak memory for information just heard • Difficulty gathering and processing thoughts in order to express them clearly • Unable to hold on to thoughts until picked to answer a question because of distraction • Losing track of an argument or stages in an experiment • The learner may have formed an answer in her/his mind and continues to think of this even though the lesson has moved on	• Ask the learner straight away before s/he forgets • Give warning of when a question is coming up • Return to learner, summarizing the argument or content so far to refresh memory and facilitate answer or comment • Use visual prompt/diagram/flow chart to 'track back'
Unable to explain what s/he should be doing in a particular activity when asked	• Sign of low working memory • Changing from one activity to another may be difficult – the learner may be competent and supported doing a task kinaesthetically but have difficulties switching to formal explanation	• Allow time to gather thoughts • Recap earlier steps • Ask 'memory buddy' to prompt • Encourage use of memory aids • Give time to express ideas as repetition can strengthen memory

Requires regular repetition of instructions	• Learner may not be able to remember instructions when given verbally • Indicates working memory overload	• Requires careful monitoring by teacher • Instructions appropriately structured to match working memory of learner (pp73-4) • Reduce number of instruction steps and possibly make instructions memorable by using actions (p76) • Structure task to include multisensory activities • Repeat instructions but ask the leaner to 'say it back' – when the leaner puts the instruction into her/his own words, understanding can be checked • Shorten instructions – reduce number of steps • Simplify language of instruction, eg in vocabulary and phrasing • From seven to eight years, learner can be encouraged to use 'rehearsal' strategies, repeating silently or aloud a small amount of verbal information to prolong short-term memory • Develop rehearsal through memorizing a simple message, saying it over and over before 'delivering' it in speech to another teacher or class

LITERACY TASKS		
Classroom behaviour	**Links to working memory**	**Strategies**
When reading, mixes up material inappropriately, eg incorrectly combines parts from two sentences rather than reading each one accurately	• Reading difficulties may arise from visual tracking problems – this should be investigated further if appropriate • Place keeping difficulties because the learner cannot remember how far s/he has got with the task (sorting information) while working on a challenging activity, eg processing writing, counting or physical actions simultaneously, leading to memory overload	• Use a tracking guide • Monitor or test eyesight and coordination if difficulties suspected • Use wider lines to avoid writing becoming obscured • Present information clearly, well spaced and without unnecessary information making the page look too 'busy' • Use different colour pen to make each line or topic area stand out
Incorrectly repeats the same response when writing, eg by writing the same word twice in a sentence	• Writing is a complex task taxing the working memory – words inserted, repeated or omitted can be a sign of working memory overload or motor memory difficulties as the learner fails to keep track of progress, especially if copying from the whiteboard • Incomplete recall may also indicate that the learner remembers what s/he wants to write at the beginning of the sentence but forgets the rest • Poor fine motor skills make writing effortful • Errors occur when focusing on handwriting while also trying to	• Encourage learner to proof read and check through once ideas are written • Encourage prior planning to help with a systematic approach for the learner to work through small steps • Offer desk copies if learner needs to copy extensive notes or objectives from the whiteboard • Avoid unnecessary copying of long date from whiteboard – use short form of date • Careful monitoring by teacher to include asking the learner what s/he intends to do next – the act of repeating important information (rehearsal) can promote storing of information in working memory (pp71-2)

	remember meaning and formulate the next sentence • Learners with poor working memory often lose track of what they have done and what has yet to be completed	
Requires support for effective use of memory aids such as useful spellings	• If memory aids are a distance from the learner's desk s/he may not use spontaneously • Learners often need demonstration, guidance and reminders of how to use and make a habit of using memory aids	• Encourage use of ICT to facilitate easy reference with word banks at a click • Familiarize learners with location of memory aids and make frequent references to a poster or wall chart • Revise memory aids occasionally to ensure they are used – have close to hand and regularly remind learner to use them (p81) • Use target high frequency words drawn from learner's own writing printed and well spaced on clear, uncluttered card – place on desk or inside exercise book • Personal spelling lists, high frequency words or topic words can be entered on Clicker 5 grids, printed and folded into a 'Toblerone' shape for easy reference • Encourage use of long-term memory class strategies, eg acronyms such as CFSS (capital letters, full stops, spellings, sense in sentences) or VCOP (vocabulary, connectives, openers, punctuation) • Ask learners to add words to a prepared list and note their own strategies for remembering (p88)
Needs regular reminders of	• Writing poses particular challenges for	• Use of three to six picture sequences with space below

each step in a written task	the learner with low working memory • If writing skills insecure, the challenge is increased	to write simple caption can aid the planning process - see writing frames information in the Primary and Secondary IDP for SLCN and dyslexia • Use learner's own drawings or photos to show a sequence of steps
Is making poor progress in literacy and maths	• Weaknesses in both subjects can be a warning sign of low memory capacity	• Track and monitor progress in both subjects • Investigate learner's strengths and weaknesses in verbal or visual-spatial short-term and working memory

STRATEGIES TO SUPPORT LEARNERS IN SECONDARY SCHOOLS

Many if not all of the suggestions in the table above will be of benefit equally in secondary school although a greater emphasis could be placed upon students adopting their own strategies to support working memory.

Learners may benefit from the following strategies.

- Small group or 1:1 sessions enabling the development of strategies that are known to make memory more efficient, such as verbal rehearsal, visual imagery, creating stories and grouping – there are a number of programmes available for schools to use to support this (see ***Resources***).

- It would be advisable, where possible, for these sessions to have strong links to curriculum content to allow the learner to see how s/he can use these strategies to support her/his learning.

- School staff should liaise closely to ensure everyone who teaches these learners are aware of the strategies that they have been trained to use.

- It would be useful to monitor the use of the strategies and discuss their effectiveness with the learner.

- Ensure written copies of any verbal information given are available as they may have difficulties remembering everything that has been said.

- If school staff agree, a dictaphone could be used to record verbal instructions and homework.

- Provide learners with brief simple instructions – they may benefit from having a written copy of instructions on their desk.

See also ***Learner strategies and training***

WORKING MEMORY AND SPECIFIC LEARNING DIFFICULTIES

Alongside the generic behaviours described above, individuals with specific learning difficulties have unique memory profiles. Knowledge of this enables specific and targeted accommodations to be made to support learning, bearing in mind the likelihood of comorbidity of learning difficulties.

DYSLEXIA AND WORKING MEMORY

Poor readers have problems with verbal short-term memory tasks, eg remembering instructions, words etc. These problems are greatly exacerbated when the learner has to process that information as well.

Learners with dyslexia show marked decrements in verbal working memory tasks. This impairment of verbal working memory is a consistent pattern in numerous studies of reading difficulties. It takes considerable working memory space to keep in mind the relevant speech sounds and concepts necessary for identifying words and understanding text, which can exceed the capacity of the dyslexic individual. For this reason, the combination of processing and remembering verbal information, rather than just remembering information, is very difficult.

Poor verbal working memory also impacts performance on phonological awareness tasks – both of these work together in written language. Both skills are needed to store phonemes of a word long enough to allow blending, to store the words in a sentence long enough to achieve meaning, and to remember the phonological and visual form of a word when writing it down.

The use of rehearsal can be problematic. The speed in which a learner rehearses information is linked to how much information the learner can hold in working memory. However, it takes a dyslexic learner much longer to repeat phrases and, as a result, s/he can run out of time to rehearse. In addition, in order for rehearsal to improve memory performance, the learner has to repeat all the information in the correct sequence, starting from the beginning of the list to the end. The dyslexic learner is unlikely to repeat information this way, and consequently s/he tends to forget all but the last bit of information heard. Generally, visual-spatial working memory is within the average range in a dyslexic learner. Converting words into images can support reading and memory, but if the information cannot easily be converted into a visual code, then the poor reader will struggle and is more likely to forget the information.

Area of difficulty	Strategies and support
General poor verbal working memory	• Build upon relative visual-spatial working memory strengths - pictures to words - visual picture strip of word sequence to reduce memory load - focus on ‘working’ part rather than the ‘memory’ part of the task
Difficulty following instructions or information in list form	• Use something concrete to help keep track of items in correct sequence on a list - avoid bullet points – use numbers instead so can keep track of how far along the list have got - use different colour for each new component of a task – learner can use colours to track progress
Verbal information processing difficulties	• Make a conscious effort to speak more slowly • Use recording devices - replay instructions at desk – can work at own pace and reduces anxieties - record own work, especially longer pieces such as essays - formulate ideas and plan work orally before writing it down • Treat ‘rehearsal’ as a memory aid which needs to be taught - teach learner what to do in step-by-step manner - ask her/him to start at the beginning of the list and say item out loud in correct sequence - if needed, encourage to move mouth quietly whilst repeating to keep it in mind
Difficulty completing tasks in timely fashion due to increased mental effort needed to process the activity	• Reduce processing load by giving a shorter activity, eg instead of 20 multiplication problems, give ten to solve – allows learner to master the concept without being frustrated that took so much effort to process each sum s/he ultimately abandoned activity altogether • Present information in a different way - vertically rather than horizontally – this supports a learner in remembering information in a sequential way • Make it easier for the learner to stay on task - highlight or mark in some way words that are difficult to read/write then come back to these later

MATHEMATICAL DISORDER AND WORKING MEMORY

Learners with a mathematical disorder can have difficulties around the following:

- deciphering mathematical symbols (eg +, ÷)
- counting principles (‘two’ stands for 2)
- solving arithmetic problems
- transposing numbers (eg 75 becomes 57)
- telling the time
- identifying left from right
- recognising patterns
- understanding quantifiers (eg know 5 is larger than 2 but not that 9 is larger than 5)
- concept of cardinality (a measure of how many elements there are in a set)
- remembering number facts – often use time consuming and inefficient strategies to support (eg counting using fingers).

However, the link between working memory and mathematical skills depends on the age of the learner as well as the type of mathematics involved. Verbal working memory plays a strong role in mathematics skills in seven year olds but is no longer significantly linked to mathematical skills once learners reach adolescence. One explanation for this is that verbal working memory plays a crucial role for basic arithmetic skills when they are young like learning arithmetic rules and retaining relevant data such as carried digits. As learner gets older, other factors such as number knowledge and strategies play a greater role.

Visual-spatial working memory problems are also linked to mathematical disorder. It has been suggested that visual-spatial memory functions as a mental blackboard to support number representation, such as place value and alignment in columns in counting and arithmetic tasks. It is also used to solve multi-digit operations and problems presented visually, including those using blocks.

Poor working memory is thought to be one explanation of mathematical disorder because it limits the ability to remember mathematical rules, from basic concepts like counting in ascending and descending order to more complicated algebraic functions. It is estimated that 50% of students with a mathematical disorder will also have reading difficulties.

Area of difficulty	Strategies and support
Poor visual-spatial working memory affecting tasks that involve numbers	• Use visual strategies to help maximize visual-spatial working memory space – eg directional arrows or cartoon symbol to support procedure of solving a problem • Consider using a number ladder rather than a number line/track – vertical representations easier to grasp in terms of more/less etc
Difficulty transcribing answers from one location to another	• Present arithmetic problems vertically not horizontally – latter requires more resources to keep the numbers in mind and carry the correct number to solve the problem • Simplify the presentation of sums to reduce processing load of trying to keep each step of the procedure in mind
Struggles to make use of concrete, visual aids available in the classroom	• Familiarize learner with the aids – learners can become overwhelmed trying to remember what teacher has said, processing that information, as well as finding a suitable aid to support learning. It is important to make the learner feel comfortable using these aids before adding on a memory activity.
Lack of mathematical knowledge and difficulty in using working memory to process information means not able to organize thoughts and plan what to do to solve the problem	• Reduce processing component by using worked out answers to help learners acquire knowledge and problem-solving skills • Support in linking worked out examples to what already know and how it fits within the larger schema of problem solving • Use worked out examples in a group setting so developing learners can explain it to those who are struggling - reduces processing difficulties - reduces anxieties around mathematics - improves reasoning skills which continue to develop up to a year later

DYSPRAXIA AND WORKING MEMORY

Poor visual-spatial memory (both short-term and working) is the trademark memory deficit in learners with dyspraxia. When compared to their peers who do not have motor difficulties, they are seven times more likely than typically developed learners to achieve very poor scores in visual-spatial memory. This is because visual-spatial memory is used to plan and control physical movement, and these skills are housed in a different part of the brain from our verbal working memory skills. In the classroom, a learner with dyspraxia will find it more difficult to complete two visual-spatial tasks at the same time, such as organizing her/his work station while visualizing the next task.

Learners with dyspraxia tend to have relative strengths in verbal short-term memory, which means they can remember a simple sequence of instructions in the classroom. However, if they have to manipulate or use that information, they can really struggle. This means that the combined task of processing and managing information that is presented both verbally (auditorily) and visually is very difficult, and visual-spatial working memory tasks are particularly difficult.

It is worth keeping in mind that because learners with dyspraxia can also have language problems, the nature of their working memory profile is crucial to establish. If they have language impairments, their verbal working memory skills may be poor as well.

Area of difficulty	**Strategies and support**
Poor visual-spatial memory	• Use visual prompts to support the learning activity or task
Struggles to make use of visual aids available in the classroom – difficulties finding relevant information and matching it to the work	• Make prompts and visual aids directly accessible • Place class schedule as images of the front cover of the relevant textbook directly on learner's desk
Difficulty in written tasks – poor fine motor skills and poor visual-spatial working memory	• Use of speech-to-text software – will allow to focus on communicating ideas rather than mechanics of writing which can be so effortful there is no memory 'space' left to remember what the learner wanted to write
Physical activity in general difficult	• Work with a buddy for things like tidying desks and putting materials away – make such paired tasks normal for class as whole to minimize isolating dyspraxic learner

ATTENTION DEFICIT HYPERACTIVIY DISORDER (ADHD) AND WORKING MEMORY

In addition to the characteristic hyperactive and impulsive behaviour, learners with ADHD also exhibit classroom behaviour typical of working memory problems. Teachers typically consider such learners to have short, poor attention spans and high levels of distractibility. They are described as forgetting what they are currently doing and things they have learned, failing to remember instructions and leaving work incomplete. In everyday classroom activities, these learners often make careless mistakes, particularly in writing, and have difficulty in solving problems. In contrast, relatively few of the learners exhibit high levels of hyperactive and impulsive behaviours. However, such problems often go undetected in the classroom or are mislabelled as a lack of motivation. As both problems with attention and working memory are linked, checklists that successfully identify learners with problems in these areas are useful to support them in the classroom.

Learners with ADHD have been widely reported to have impairments in these key executive functions:

- inhibition
- shifting
- planning actions.

These learners also have difficulty in a related executive function skill that requires them to focus their attention on a single task, known as ‘sustained attention’.

However, learners with ADHD do not have difficulties with short-term memory. In a testing situation, they can recall digits, words, instructions and spatial locations at the same rate as their peers. In the classroom, they can usually remember what they have been told and repeat it back. The problem arises when they actually have to manage or manipulate that information, in both the verbal and visual domains. They struggle when they have to hold information in mind and use that to guide their behaviour towards tasks or goals. Poor working memory affects all areas of learning throughout their academic career. Research suggests that visual-spatial working memory deficits are such a prominent feature of learners with ADHD that this can be used as an additional diagnostic tool to identify them in the classroom.

Area of difficulty	Strategies and support
Lose information because cannot simultaneously maintain attention and use working memory while teacher is speaking	• If possible, keep list of instructions short - take the average limit in terms of how many things can be remembered at the learner's age and give one or two items less to reduce processing load • Shorten study periods to reduce memory loss of information in the middle section of an activity/task/instruction - if class are asked to spend 20 minutes reading, ask the ADHD learner to spend eight minutes reading - build in a three minute break before coming back to complete the reading - use a visual timer so that the learner has a clear idea of how much time they have to spend
Difficulty keeping track of time or managing own schedule	• Provide something explicit that they can see, eg egg timer • Break tasks up in the middle • Specify timed bursts of effort rather than a prolonged period of time on an activity
Finds process of rehearsal problematic	• Rehearse information sporadically rather than at a set time each week • Reinforce information that is being revised by testing periodically, eg using quick quizzes
Difficult to motivate and keep on task	• Offer rewards frequently and sporadically – avoid 'delayed gratification'
Abandons activities when they become too effortful	• Reduce effort expended in trying to remember information • Utilize distractability – encourage to look around and be aware of environment - ask them to think of where they were when they learnt the information they are trying to remember, eg what chair were they sitting on? What time of day was it? • Remembering aspects of physical environment can help trigger memory
Difficulty maintaining focus when listening	• Harness her/his energy! - ask her/him to perform a physical activity while listening to information, eg kicking an imaginary football while listening to a story about sport - actions do not necessarily need to match information – ask to perform a series of actions whilst listening to information, eg stand up, cross your arms, hands on head then ask to recall the information specifically connected with each action

AUTISTIC SPECTRUM DISORDER (ASD) AND WORKING MEMORY

Learners with ASD have difficulties with social communication, with imagination and with flexibility of thought. Unusual responses to sensory stimuli are also common, and can be a source of distraction and discomfort in the classroom. The social and physical environment of school can therefore be a challenging place.

Research into memory suggests that learners with ASD often have problems with verbal short-term memory. This weakness can impact on language learning. It is less clear whether or not they have problems with working memory as such, but performance on tasks involving working memory in the classroom is likely to be affected by the underlying strengths and weaknesses these learners have. Some learners with ASD can display verbal working memory weaknesses when they have to visually scan material to pick out the relevant information. In the classroom, some learners with ASD might appear to struggle with certain memory-heavy activities. However, this may be connected to their difficulty in knowing what they should focus on, rather than a working memory deficit per se.

It seems that the working memory profile of an autistic savant is much like an individual with ASD without special abilities in general, expect in the area of their expertise. This can sometimes improve memory skills in related areas. This enhanced memory performance often far exceeds what is expected of them when looking at IQ levels/underlying abilities.

Learners with ASD tend to have visual strengths and respond well to structure and routine.

Area of difficulty	**Strategies and support**
Difficulty coping with change	• Have a consistent approach within school and liaise closely with parents about how situations are dealt with • Keep to a structured routine • Show what will be happening throughout the day visually, eg on a visual timetable • A timer may help to signify when one activity ends and the next starts • Prepare for (expected) changes in advance • Introduce the idea of a nice change through a ‘surprise’ card on the visual timetable - sometimes change will not be predictable in advance but showing what is now happening visually should offer reassurance

Levels of emotional arousal and stress build up quickly and present barriers to learning – the ASD learner may have difficulty recognising, communicating and dealing with these feelings	• Teach understanding of basic emotions and how they are linked to situations • Introduce ways to communicate feelings, eg through emotions cards • Have an identified space the learner can go to if stressed and have an agreed way of communicating that s/he needs to do this (again a cue card may be helpful)
Verbal comprehension difficulties	• Use clear, concise language and be alert to potential problems with the learner taking language too literally • Give time to process verbal information (ten second rule) • Use visual information (pictures, photos etc) to back up verbal information • Present tasks visually, in a series of steps • Be alert to problems of hyperlexia – good reading accuracy but poor comprehension
Problems with social skills and social understanding	• Teach social skills and social understanding - approaches such as social stories and *The Social Use of Language* programme can be very helpful • Avoid social and academic challenges within the same task
Difficulty focusing attention, maintaining attention and shifting attention – having a 'monotropic' focus of attention and finding sharing attention between two or more sources of information difficult	• In a classroom it may be difficult to shift attention from the teacher, to the interactive whiteboard and then back to the teacher who they will need to track as they move around the classroom - consider carefully the best way to convey information to the learner • Computer aided learning may be an option
Unusual responses to sensory stimuli that can cause great distress/ distraction	• See *Sensory Audit for Classrooms* and *Sensory Checklist for Pupils* in the Primary and Secondary IDP for ASD
Problems generalizing skills to a new situation	• Teach new skills in a range of different situations • Mind maps can help learners to see connections

LEARNER STRATEGIES AND TRAINING

META-AWARENESS SKILLS

Meta-awareness skills can promote self-regulated learners who know how to use their own cognitive tools to support memory skills. The following steps can help to develop this meta-awareness.

- **WHAT did you do?** Encourage the learners to try to look 'inside' their working memory. When they can say what they did, move on to the next question.
- **WHY did you do that?** This will help learners to be aware of their own memory strengths and weaknesses, as well as their preferences for learning. Now ask the next question.
- **HOW will it help you?** It is important to get learners to articulate why they chose their strategy so that they can begin to see the connection between what they are doing and the lesson. The learner may need help in refining this strategy if it is overly complicated but it should be the learner's own strategy. This leads on to the final question.
- **WHEN can you use it again?** Learners with low working memory often view strategies as a one-off activity and do not think of applying them to a different problem. If the learner is having trouble generating examples of when s/he can apply the strategy again, ask her/him to pick out a key feature in the problem and identify another activity that has the same feature. Now ask her/him if the strategy would also work for that new activity and why it would be suitable.

PROMPTING DEEPER UNDERSTANDING

The following are suggestions to encourage the learner with low working memory to move from a superficial engagement to a deeper understanding of the material.

- **Think aloud.** When working on her/his own, ask the learner to think about three main points in what s/he has read instead of just rereading the text. This will encourage her/him to process the information instead of just skimming over the text.
- **Talk aloud.** Ask learners to work in pairs and discuss what they read. Studies have found that when learners talk about what they have read, they remember more information than if they just read a list of summary points.

CLASSIFYING AND CATEGORIZING

One way to support learning is to use cues to help learners remember. Studies have found that when learners use category cues they are twice as likely to remember all the words associated with that category: eg the vocabulary *shoes, hat, belt, shirt, trousers, chair, table, desk, lamp*, can be categorized into ‘clothes’ and ‘furniture’. It is important to note that learners younger than seven years do not immediately form semantic associations and so will need time spent making these connections evident to them.

This technique is a useful and necessary preliminary to developing the independent use of the skill of mind mapping to remember, revise and review information.

MAKING THE LINKS

This involves making a bridge from new information to existing knowledge in long-term memory. One way of doing this is to create ‘memory stories’ or ‘memory movies’. Here key facts and knowledge are woven into a narrative the learner creates.

Another similar technique is to create a ‘memory journey’ which works in much the same way, with the learner using imagination to ‘see’ the things s/he wants to remember on a familiar journey.

REVIEWING LEARNED INFORMATION

This technique can be found in *Mastering Memory* (Mitchell, 2001) and is based on the work of Tony Buzan. If information is learned effectively in the first place, the reviewing process relies on the principle that it is better to go over a piece of information while it is still easily recalled, rather than allowing it to be forgotten. Once forgotten, it requires considerable effort to relearn. In the reviewing process, information should be reviewed five minutes after it was learned. This checks that the skill or information was really learned and increases the chance of remembering it for about a day. The intervals at which subsequent reviews should be performed are:

- the next day
- one week later
- one month later
- three months later
- six months later
- one year later

RESOURCES

SOFTWARE

Name	Publisher/Cost [correct as of Feb 2011 excluding VAT unless stated]	Brief description	School stage	
			1ary	2ary
Mastering Memory	CALSC PO Box 621, Sutton, Surrey SM1 2ZS Tel: 01797 361318 www.masteringmemory.co.uk From £82.50 for children's version From £150.00 for complete version	*Mastering Memory* is a metacognitive memory training program, developed by a speech and language therapist and dyslexia teacher. It is a tool which under guidance of a learning support assistant, therapist or teacher can be used to facilitate strategies to improve visual and/or auditory short-term memory; enhance the capacity to transfer information to long-term memory; and transfer new memory skills to other situations. It can also be used as a dynamic assessment tool by an experienced practitioner.	√	√
Timely Reminders and Time to Revise	CALSC As above Tel: 01797 361318 www.timelyreminders.co.uk £12.95 download (including VAT) £27.95 CD-ROM (including VAT)	*Timely Reminders* helps users to remember information by structuring the revision process. By storing key facts in the program's database, along with questions and answers related to the facts, the user can build her/his own personalized revision timetable. Once the student has learned the key facts thoroughly, s/he can focus mental energy on applying them instead of struggling to remember them. Can be used independently at home by students.	√	√
Memory	Lucid Research Limited	*Memory Booster* is a computer program that is effective	√	√

Booster	3 Spencer Street, Beverley, East Yorkshire HU17 9EG Tel: 01482 882121 www.memorybooster.co.uk Home edition £19.00 (CD or download) Standard edition from £37.00 for single and £195.00 for network	even without parental or teacher supervision. Memory Booster teaches flexible memory strategies and takes the child through enjoyable exercises that develop and extend learning skills. Memory Booster is adaptive, so the activities are automatically adjusted to the abilities of each individual child. Learning progresses at the pace that is best suited to their needs. However, parents or teachers can configure the program to make it easier or harder if this should be necessary. Although aimed at primary school children, also effective for secondary students especially those with special educational needs including dyslexia and ADHD.		
BrainBooster	Nessy 10 Upper Belgrave Road, Clifton, Bristol BS8 2XH Tel: 0117 923 9777 www.nessy.com/brainbooster £42.50 for single user	*BrainBooster* is an academic study skills programme aimed at students between the ages of 13 and 18. Designed to improve memory skills and provide study strategies for individual students, it can be used independently to improve grades and to help pass exams.		√
iMindMap V4 Elements	ThinkBuzan Regus House, Falcon Drive, Cardiff CF10 4RU Tel: 0207 117 7173 www.thinkburzan.com/uk/products/imindmap	*iMindMap* is the official mind map software from the inventor of mind mapping Tony Buzan. It is the only tool that fully and accurately replicates his process. This technique is claimed to help students become more creative, enabling better decision making and problem-solving in less time.	√	√

	From £41.70 (digital download available)			
2Connect	2Simple Software Enterprise House, 2 The Crest, Hendon, London NW4 2HN Tel: 0208 203 1781 www.2simple.com From £39 for a single licence	*2Connect* is a concept mapping and planning tool which will help children organise their thoughts individually or collaboratively on a network. It is a simple way to put ideas on a page and has the following features: • Concept Map – add words, sound and images • Sound Recorder – record and play back own voice to help support and guide writing • Word Bank – link to any writing program, click on a word and it is inserted into the text • Planning Tool – can export into 2Create and the multimedia project is created automatically.	√	

WEB-BASED				
Jungle Memory	Memosyne Ltd www.junglememory.com From $49.99 for 8 week online subscription	*Jungle Memory* is an online training program designed to dramatically improve working memory and grades. The program directly trains working memory to stimulate how children from ages six to 16 learn. The end result is claimed to be improved classroom performance and higher grades. It can be self-directed at home or bought in for school use. Endorsed by Tracy Packiam Alloway.	√	√
iKonMap	www.ikonmap.com Free	Concept mapping tool with facility to save and print out maps.	√	√

BOOKS AND OTHER RESOURCES

Name	Publisher details	Brief description		
Student Organiser Pack	CALSC PO Box 621, Sutton, Surrey SM1 2ZS Tel: 01797 361318 www.studentorganiserpack.co.uk £15.26 download (including VAT)	The *Student Organiser Pack* helps students from ages nine to 19 years old to be more organised, make more efficient use of their time and to achieve their aims more successfully. It helps students to learn the skills of time management. A questionnaire enables students to find their own strengths and weaknesses. Once downloaded, teachers may photocopy as many times as needed.	√	√
The Buzan Study Skills Handbook by Tony Burzan	BBC Active (2006) From around £7.50 on Amazon	'In Buzan's Study Skills, three of Tony Buzan's most enduring subjects - Memory, Speed Reading and Mind Mapping - are combined in a single guide to enable students to make the most of their studies. Improve grades, make more effective notes, plan better essays, make associations between different subjects, have at-a-glance notes and records for easy revision, learn how to use effective memory techniques to recall information and increase reading speeds up to 1000 words per minute. The perfect guide for any student who wants to get more from their studies, improve their brainpower, consolidate their concentration and learn from the master of memory - Tony Buzan.' (From publisher's blurb)	√	√
Memory Trainers by Fil Came and	Available as a pdf download for	'Practical techniques, games and systems to improve memory skills. This book has been developed with	√	√

Gina Cooke	£8.25 from website shop.learning-works.org.uk	pupils in both primary and secondary schools. By following a clearly defined programme improvements have been observed in terms of confidence, self belief, motivation and concentration. The activities and games aim to improve self-esteem by showing what can be learned, there are many different ways to learn, and learning can be fun!' (From publisher's blurb)		
Study Skills: A Multisensory Guide by Jenni Beard	Crossbow Education Ltd Tollgate Court Business Centre, Tollgate Drive, Stafford ST16 3HS Tel: 0845 269 7272 www.crossboweducation.com £4.50	'*Study Skills: A Multisensory Guide* is user friendly and gives practical help on study skills that is needed for students facing the pressures of exam revision from 15-16 years on through college and university. It covers study skills approaches for all learning styles, including essential topics such as mind mapping, proof-reading, relaxation techniques, timetabling your study programme. There is a centre spread revision timetable that can be photocopied.' (From publisher's blurb)		√
Memory Skills Chute Cards	Smart Kids 5 Station Road, Hungerford, Berkshire RG17 0DY Tel: 01488 644 644 www.smartkids2.co.uk Smart Chute £15.99/Chute Cards £4.99 each/All four levels £18.99	'When posting the card in the Smart Chute, the learner can say the answer and the correct answer then miraculously appears at the bottom. The Chute is made out of solid plastic and its durability will withstand maximum classroom use. A perfect resource for individual work, the Smart Chute is beneficial for all children but especially those with specific learning difficulties or those learning English as an additional language. Memory Skills Chute Cards have been designed to develop visual memory skills and improve working memory. Available at four levels.' (From catalogue description)	√	

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